Business Question and Visualization Report

Date	29 October 2026
Team ID	N/A - individual project; no Team ID provided
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	5 Marks

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

Business Questions and Visualisation

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations in Tableau. Well-framed questions help in identifying key metrics, selecting relevant data, and building visualisation that provide insights.

Project-specific business questions and Tableau visualizations

Nine business questions were mapped to the implemented Tableau views. The dataset is synthetic; insights describe this sample only.

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- o See the linked published Tableau dashboard. Screenshots are maintained separately; this document includes the chart inventory and evidence link.

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Completed: 9 distinct visualizations are listed below.

#	Business Question	Visualization / Encoding	Key interpretation / caveat
1	How is recorded annual sales volume distributed across vehicle categories?	Horizontal bar chart: SUM(Annual Sales Units) by Category, sorted descending.	Two-Wheeler Scooter is highest in the cleaned sample at 175,389,726 recorded sales units; figures describe synthetic records only.
2	Which manufacturers have the largest recorded sales totals?	Treemap: Manufacturer sized by SUM(Annual Sales Units).	AmpereWorks leads the manufacturer totals in this sample at 9,504,546; names are fictional.
3	How many customer-rating records are represented by each EV category?	Packed-bubble chart: Category with COUNT(Customer Rating); bubble size represents record count, not rating score.	Bubble area compares the number of records in each category; it must not be interpreted as average satisfaction.
4	How do annual sales units vary by model year cohort?	Line chart: SUM(Annual Sales Units) by Model Year (2018-2026).	This compares model-year cohorts in the dataset and is not a historical calendar-time industry trend.
5	How do battery technology types compare in recorded sales contribution?	Pie chart: Battery Type sized/angled by SUM(Annual Sales Units).	NMC and LFP have the largest recorded totals in this synthetic dataset.
6	Which manufacturing countries carry the largest recorded totals?	Filled map: Manufacturing Country colored by SUM(Annual Sales Units).	India and China are highest in this sample; values must not be presented as verified real-world production.
7	How is five-year battery degradation associated with vehicle category and battery type?	Heat map: Category by Battery Type, color by AVG(Battery Degradation Pct 5Yr).	The matrix supports category/chemistry comparison; interpret differences descriptively rather than causally.
8	How does driving range compare with battery capacity?	Scatter plot: AVG(Battery Capacity Kwh) vs AVG(Range Km), color by Battery Type.	The plot visualizes range and capacity together; overlapping points and a fitted line should not imply causality.
9	How does average price vary across vehicle categories?	Dot plot: AVG(Price Usd) by Category.	The view helps compare category-level price positioning within the synthetic records.

Dashboard URL: https://public.tableau.com/shared/Q27K39PSC?:display_count=n&:origin=viz_share_link

Story URL: https://public.tableau.com/views/EV_Market_Intelligence/Story1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Note: The customer-rating bubble view encodes record counts, not average rating. No formal user research was conducted. Use the published Tableau views for actual interaction evidence.